

Regulatory Separation and Pharmaceutical Innovation: Evidence from China's MAH Reform

Abstract

This paper develops a partial-equilibrium mechanism model of how China's Marketing Authorization Holder (MAH) system changes the value of advancing viable drug projects and their commercialization organization. Developers choose one common project-development / advancement intensity before project characteristics and organizational routes are realized. MAH makes retained entrusted manufacturing legally available through one institutional wedge, while route choice remains deterministic and the developer retains authorization-holder responsibility. Internal and entrusted production are derived from distinct technologies, and the price of qualified manufacturing services is endogenous. Direct route-value gains occur for project-developer pairs where the new route improves on the best old option. These gains raise common advancement and can be attenuated by service scarcity; common advancement may also raise outcomes in a class with no direct route-value gain. The baseline does not treat MAH as a demand, scientific-productivity, downstream-success, or patenting shock, and it imposes no original-versus-incremental response ranking.

Keywords: MAH; project development; commercialization frictions; entrusted production; route choice.

JEL codes: D22; L65; O31; O38; I18.

1 Introduction

China's MAH system separates the right to hold marketing authorization from a strict requirement of in-house production. This institutional change is potentially important for drug development because many research-originating firms can generate valuable projects but lack complete manufacturing capacity. The central mechanism studied here is therefore not demand expansion or scientific productivity. It is an expected-commercialization-value channel: when a firm anticipates that an advancing project can later use qualified external manufacturing while the firm retains authorization, the expected value of advancing that project may increase.

The mechanism builds on two established ideas. First, pharmaceutical innovation responds to expected downstream rewards ([Schmookler, 1966](#); [Acemoglu and Linn, 2004](#); [Finkelstein, 2004](#); [Dubois et al., 2015](#); [Budish et al., 2015](#)). Second, an innovator’s ability to appropriate returns depends on complementary assets and commercialization strategy ([Teece, 1986](#); [Arora et al., 2001](#); [Gans and Stern, 2003](#)). MAH fits this second logic because it changes how an innovator may access manufacturing capability while retaining the authorization asset.

Recent evidence on China’s pharmaceutical reforms helps separate this route mechanism from two nearby channels. [Jia et al. \(2023\)](#) study the 2015 drug-review reform and emphasize approval delay, review capacity, and the composition of firms and products entering the pipeline. [Barwick et al. \(2026\)](#) study the National Reimbursement Drug List and emphasize demand and market-size incentives. Those channels can coexist with MAH, but they are not the MAH mechanism in this paper. The baseline holds scientific productivity and the market-return environment fixed and asks whether a regulated holder–producer separation route changes expected commercialization value.

Closely related evidence also shows why project advancement and upstream patenting must remain distinct. [Gu \(2024\)](#) finds that allowing production outsourcing raises clinical-trial registrations among innovators without production facilities, while patent applications fall among firms with fewer financial resources. The paper also reports heterogeneous responses across development stages and drug classes. These findings motivate empirical heterogeneity tests, but they do not make patenting a baseline outcome here, do not impose an original-versus-incremental ranking, and do not import a financial-constraint mechanism into the model.

For a development-economics audience, the relevant bottleneck is a regulated complementary asset. MAH can relax a manufacturing bottleneck for research-oriented firms while preserving holder responsibility. Scarcity in qualified production support can still attenuate the effect.

The model is deliberately narrow. It is not a full dynamic structural model of the pharmaceutical industry. It does not solve product-market clearing, an input market for project advancement, free entry, exit, or an invariant firm distribution. The baseline closes only the market directly tied to the MAH mechanism: qualified manufacturing services. Reform-induced demand for retained entrusted production can raise the equilibrium service price p_m^* , so the model does not mechanically imply a positive effect for every project or developer.

The contribution is to organize the mechanism in a way that is internally consistent with feasible data. The theory separates project advancement from planning-stage project arrival, deterministic organizational assignment, observed holder–producer separation, and realized products. The empirical role of the model is correspondingly modest:

it specifies empirically interpretable interfaces for project-level organization, retained outcomes, and service scarcity without claiming that approval-side data separately identify every primitive friction.

The model is presented in two layers. The main text gives the shortest complete version needed for the paper’s argument. The technical appendix records primitives, payoff accounting, pricing derivations, deterministic sorting, project advancement, qualified-capacity equilibrium, comparative statics, observed outcomes, and explicitly separated extensions.

2 Institutional Background

The relevant institutional point is timing. Under China’s drug-registration framework, a firm need not already be the final commercial holder when it begins to advance a viable project. The economic choice is forward-looking: a firm invests today while anticipating the manufacturing and commercialization environment it will face if a project advances. Production support and manufacturing readiness must be planned before final approval is observed. After obtaining a Drug Approval License, the applicant becomes the MAH ([National Medical Products Administration, 2022a](#)). The model therefore treats MAH as changing the anticipated availability and value of a future organizational route, not current scientific productivity or downstream success.

The MAH reform creates a regulated holder-producer separation route. The 2016 pilot allowed eligible applicants in selected regions to become holders under the pilot arrangement ([General Office of the State Council of the People’s Republic of China, 2016](#); [China Food and Drug Administration, 2016](#)). The later legal framework consolidated the principle that an MAH may manufacture by itself or entrust production to a qualified manufacturer ([National Medical Products Administration, 2019](#)). Subsequent NMPA rules require holder-side quality responsibility, assessment of the entrusted manufacturer, quality and entrustment agreements, and ongoing supervision ([National Medical Products Administration, 2022c,b, 2023](#)).

These rules imply two modeling restrictions. First, entrusted production is not anonymous outsourcing or transfer. It is a regulated retained route in which authorization remains with the developer while production is performed by a qualified external manufacturer. Second, MAH does not eliminate holder responsibility. The entrusted payoff therefore retains a holder-side burden μ_E , while the institutional wedge $\tau_E(M)$ governs effective route availability. The downstream realization probability $s(q)$ is route independent and invariant to MAH in the baseline.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined research capability $a_i > 0$ and internal manufacturing capability $k_i > 0$, distributed according to $H(a, k)$. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q, m) = \sum_{g \in \{O, \text{Inc}\}} \rho_g F_g(q, m), \quad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \quad \tau_E(1) = \bar{\tau}_E < +\infty, \quad \bar{\tau}_E \geq 0. \quad (1)$$

The available routes are internal manufacturing I , retained entrusted manufacturing E , non-retained transfer T , and abandonment or delay A . Under E , the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as project-development / advancement intensity. It advances viable drug projects toward commercialization-relevant route planning and excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of viable projects reaching route planning is

$$\lambda_i^{\text{plan}} = a_i x_i. \quad (2)$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price. Second, they choose x_i . Third, planning-stage projects draw (q, m) and choose a route. Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. Upstream research and patent generation may help determine the predetermined a_i , but they are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p; q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c$$

and the gross operating present value

$$R(q, c) = \frac{Aq}{1 - \beta\varphi} \frac{(\varepsilon - 1)^{\varepsilon-1}}{\varepsilon^\varepsilon} c^{1-\varepsilon}, \quad \beta \in (0, 1), \quad \varphi \in [0, 1). \quad (3)$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$ on the feasible domain. Internal production is unavailable when $k_i < \underline{k}(m)$, represented by $F_I(m, k_i) = +\infty$. On the feasible domain its route value is

$$W_i^I(q, m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \quad (4)$$

Below the feasibility threshold define $W_i^I = -\infty$ directly; c_I need not be defined there. Cost derivatives apply only to the feasible interior.

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m ,

$$\begin{aligned} W_i^E(q, m; M, p_m) &= s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ &\quad - \mu_E - \tau_E(M). \end{aligned} \quad (5)$$

Both marginal costs are strictly positive on their domains; setup costs are nonnegative when finite. The capacity payment $p_m b(m)$ is excluded from c_E and R . Each monetary cost enters once. The transfer and abandonment values are

$$W^T(q, m) = T(q, m), \quad W^A = 0. \quad (6)$$

3.3 Organization and project advancement

At a fixed p_m , the project chooses deterministically:

$$\begin{aligned} W_i(q, m; M, p_m) &= \max\{W_i^I, W_i^E, W^T, W^A\}, \\ r_i^*(q, m; M, p_m) &\in \arg \max_{r \in \{I, E, T, A\}} W_i^r. \end{aligned} \quad (7)$$

Here $W_i^T = W^T$ and $W_i^A = W^A$. We assume measurable values and zero probability of maximizing ties at each candidate price and regime for almost every developer; continuity of heterogeneity alone is not sufficient. A measurable tie rule completes the definition on null sets.

The internal-minus-entrusted gap is

$$\begin{aligned} \Delta_{IE}(k_i) = & s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ & + F_E(m) + p_m b(m) + \mu_E + \tau_E(M). \end{aligned} \quad (8)$$

Holding (q, m, M, p_m) fixed with finite τ_E , on the internally feasible interior,

$$\Delta_{IE,k} = s(q)R_c(q, c_I) c_{I,k} - F_{I,k} > 0.$$

When the continuous gap has a negative lower-end limit and a positive upper-end limit, a unique conditional cutoff $k^*(q, m; p_m, M)$ satisfies $\Delta_{IE}(k^*) = 0$. Conditional on I and E dominating T and A , developers below the cutoff select E , while those above it select I . Transfer and abandonment remain valid outside options.

Expected optimized value per planning-stage project is

$$\Omega_i(M, p_m) = \int W_i(q, m; M, p_m) dF(q, m). \quad (9)$$

Developer i solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1 + \nu} x_i^{1+\nu} \right\}, \quad \kappa > 0, \quad \nu > 0. \quad (10)$$

Strict concavity gives the unique solution

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}, \quad (11)$$

including $x_i^* = 0$ when $\Omega_i = 0$. The binary reform affects x_i^* only through Ω_i . At a fixed p_m , its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j , with efficiency z_j , chooses capacity $s_j \geq 0$:

$$\max_{s_j \geq 0} \{p_m s_j - \Psi(s_j; z_j)\}. \quad (12)$$

Assume $\Psi(0; z) = \Psi_s(0; z) = 0$, $\Psi_{ss} > 0$, and marginal cost diverges with capacity. At a positive price, capacity is uniquely determined by $p_m = \Psi_s(s_j^*; z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m, z) dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m; M) = \int b(m) \mathbf{1}\{r_i^*(q, m; M, p_m) = E\} dF(q, m). \quad (13)$$

Study-related and total demand are

$$\begin{aligned} D_m^{\text{MAH}}(p_m; M) &= \int a_i x_i^*(M, p_m) \chi_i^E(p_m; M) dH(a, k), \\ D_m(p_m; M) &= D_m^B(p_m) + D_m^{\text{MAH}}(p_m; M). \end{aligned} \quad (14)$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. The no-maximizing-ties condition, continuity of finite route values in price, and the integrable value/capacity envelopes stated in the appendix make aggregate demand continuous even though individual indicators can jump. Supplier responses are integrable on compact price intervals. Background demand is finite, continuous, weakly decreasing, positive at zero, and tends to zero at infinity.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \quad (15)$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

The baseline partial equilibrium contains exactly

$$\{p_m^*, x_i^*, r_i^*(q, m), s_j^*\}_{i,j}. \quad (16)$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Main predictions

Organizational sorting. Conditional on I and E dominating the outside options, low internal manufacturing capability selects entrusted production and high capability selects internal production. Lower entrusted-route friction expands the entrusted region;

a higher fixed CMO price contracts it.

MAH-relevant projects and advancement. The reform has a zero effect whenever the newly feasible E value does not exceed the best old route. Otherwise it raises expected project value and, at fixed service price, raises common project advancement. Holding the initial surplus fixed, the response increases with the value gain. Across developers, higher a_i scales a given surplus response upward; a larger response at lower k_i additionally uses $\nu \geq 1$ and the feasibility-boundary continuity conditions in the appendix. No general capability ranking is imposed for $0 < \nu < 1$. MAH does not change a_i , project quality, or the advancement-cost function.

CMO equilibrium and scarcity. The qualified-capacity price is unique. Reform-induced entrusted demand can raise this price, weakly reducing the direct fixed-price gains in project value and advancement. Fixed-price comparisons and equilibrium-price comparisons are therefore distinct.

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF(q, m). \quad (17)$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \text{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$, with the same realization kernel $s(q)$. A zero class-specific value gain alone does not imply a zero outcome effect: the common advancement response also scales that class’s existing retained outcomes. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.

4 Empirical Predictions and Mapping Boundaries

The deterministic model yields three conditional predictions. First, among projects for which internal and entrusted production dominate transfer and abandonment, developers

with weaker internal manufacturing capability are more likely to select retained entrusted production. Second, MAH raises project advancement only when the new entrusted value exceeds the best pre-reform option; the response can be zero for many developers. Third, reform-induced demand for qualified manufacturing services can raise p_m^* , attenuating fixed-price gains in expected project value and advancement. These predictions concern commercialization organization and project advancement, not patent generation or scientific productivity.

The empirical distinction is between ex ante advancement and downstream observations. Product-level holder–manufacturer separation can proxy route E only when holder identity, manufacturer identity, product identity, and effective date are aligned. Approval or launch records are realized-product outcomes, not direct observations of x_i . Clinical milestones may proxy project advancement only when their status and timing are measured consistently. Patent applications are a separate patenting margin and are not a baseline endogenous outcome.

Table 1: Model Objects and Measurable Empirical Interfaces

Model object	Candidate interface	Claim boundary
Predetermined capability a_i Advancement x_i and $a_i x_i$	Pre-policy development history or other validated capability measure Consistently measured development-stage activity and planning-stage project events	Does not separately identify a_i , and no policy shift in a_i is allowed. x_i advances viable projects toward route planning; it excludes patent-generating research and is not an approval count.
Route $r_i^* = E$	Product-level holder differs from manufacturer, with time-consistent identifiers	The observed split is downstream evidence on retained organizational assignment, not the policy treatment itself.
Retained outcome Y_i^{ret}	Approval or launch with developer/holder identity	Combines advancement, project mix, deterministic route assignment, and $s(q)$; it does not identify these primitives separately.
CMO scarcity p_m^*	Service price, or consistently measured capacity, utilization, density, waiting-time, or participation proxy	A nonprice proxy does not identify the price or the supply and demand schedules separately.
Novelty class g	Versioned regulatory classification O or Inc	The classifier creates no x_{ig} and imposes no response ranking.

The timing restriction can be summarized as

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*, \quad (18)$$

followed by

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \end{aligned} \quad (19)$$

Observed separation cannot be placed before or treated as a cause of the ex ante x_i choice. The original and incremental classes use the same x_i and the mixture $\rho_g F_g$; either class may have a zero or larger response.

These entries are interfaces for future validated data. No data set is asserted here to be available, complete, or identifying. Before use, a source must pass provenance, coverage, key-uniqueness, unit, timing, and construct-validity checks. Entry, route-specific realization, smooth route choice, and research-versus-development allocation remain separate extensions rather than baseline calibration objects.

5 Conclusion

MAH changes expected project value by making retained entrusted manufacturing available while preserving authorization-holder responsibility. The route is most relevant when a developer can advance valuable projects but lacks project-compatible internal manufacturing capability. Its equilibrium value also depends on qualified-service scarcity: reform-induced entrusted demand can raise p_m^* and attenuate the fixed-price gain. Common project advancement rises only for developers with a positive expected route-value gain. A class may benefit through common advancement even without its own route-value gain, but no strict effect or ranking is required for every novelty class. Observed holder–producer separation and realized products occur after advancement and organizational assignment. Patent applications and upstream scientific research are outside the baseline, and the model makes no welfare claim. This deliberately narrow partial-equilibrium structure keeps the paper focused on commercialization organization, manufacturing mismatch, retained authorization, and CMO scarcity.

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